

4.34)

Besümm Rms Wärdet AV $v(t) = 35 + 63 \cos(215t)$

Lösung:
DEF. $v_{rms}^2 = \frac{1}{T} \int_0^T |v(t)|^2 dt$

vi RZ

$$v_{rms}^2 = \frac{\omega}{2\pi} \int_0^{2\pi/\omega} (35 + 63 \cos(\omega t))^2 dt =$$

$$= \frac{\omega}{2\pi} \int_0^{2\pi/\omega} (35^2 + 4410 \cos(\omega t) + 3969 \cos^2(\omega t)) dt =$$

$$= \frac{\omega}{2\pi} 35^2 \frac{2\pi}{\omega} + 4410 \frac{\omega}{2\pi} \frac{1}{\omega} (\sin(\omega \frac{2\pi}{\omega}) - 0) +$$

$= 0$

$$+ 3969 \frac{\omega}{2\pi} \int_0^{2\pi/\omega} \frac{1}{2} (1 - \sin(2\omega t)) dt =$$

$$= 35^2 + 3969 \frac{\omega}{2\pi} \frac{1}{2} \left[\frac{2\pi}{\omega} + \frac{1}{2\omega} \cos(2\omega \frac{2\pi}{\omega}) - \frac{1}{2\omega} \right] =$$

$$= 35^2 + 3969 \frac{1}{2} = \underline{\underline{56.65 \text{ VOLT}}}$$

SVAR:

$$v_{rms} = 56.65 \text{ V}$$